

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

1. First , calculate the heat needed to raise the temperature of 1 liter (or 1 0 0 0 grams) of water from 30°C to 60°C :

- Heat capacity of water = 1 cal/g°C
- Mass of water = 1 0 0 0 grams
- Change in temperature = 60°C - 30°C = 30°C
- Heat required = mass × specific heat × temperature change = 1 0 0 0 × 30 × 1 = 3 0 0 0 0 calories
- Convert calories to kilocalories : 3 0 0 0 0 cal × 0 . 0 0 1 = 30 Kcal

2. Then , calculate the heat needed to heat a 1 kg (or 1 0 0 0 grams) block of aluminum from 30°C to 60°C :

- Heat capacity of aluminum = . 2 1 5 cal/g°C
- Mass of aluminum = 1 0 0 0 grams
- Change in temperature = 60°C - 30°C = 30°C
- Heat required = 1 0 0 0 × . 2 1 5 × 30 = 6 4 5 0 calories
- Convert calories to kilocalories : 6 4 5 0 cal × 0 . 0 0 1 = 6 . 4 5 Kcal

3. Now , find the sum of the two heats :

- Total heat required = 30 Kcal + 6 . 4 5 Kcal = 36 . 4 5 Kcal

4. The closest answer choice to 36 . 4 5 Kcal is G . 30 Kcal , 6 . 4 5 Kcal , which is the closest to the calculated value .

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<answer>G . 30 Kcal , 6 . 4 5 Kcal</answer>